

Before you walk in

Open kiranic.com/lecture/live-monitor.html *first* — the trend needs ~40 sim-days (5–8 real min) before it means anything. Then open kiranic.com/lecture/bearing-lab.html in a second tab. Local HTML copies on a USB stick as WiFi backup. Deck open. Script printed.

Calibration moment — slide 4, before anything else: “Show of hands — who has finished Digital Signal Processing?”

3rd years won't have; 4th years will. If mostly 3rd year, add **5 min** to slide 15 (envelope-from-scratch), Q&A shrinks. On slide 21 (MPC): 4th years — “you're doing this now.” 3rd years — “this is what Sem VII will teach you.”

THE 34 SLIDES, MAPPED

SLIDES	TIME	TITLE	WHAT YOU DO	BROWSER
1–5	0:00–0:05	Open	Cold open (thermocouple/strain-gauge show of hands). Disclaimer: never commissioned a control loop. Agenda + DSP show of hands	monitor hidden
6–10	0:05–0:13	Part 1 · The stack	Move briskly. Slide 9: kill the microsecond-AI myth (off by 10,000×). Slide 10 is the rule	—
11–13	0:13–0:17	Part 2 setup	Derive BPFO = 107.4 Hz out loud. Numbers before measurement — this is the whole point	—
14–15	0:17–0:23	Why FFT fails + envelope	Slide 14: shaft line is 170× the defect. Slide 15: 4-min build (+5 if 3rd-year). Delete negative freqs → transform back → magnitude. Land on: “you already designed this filter in DSP, you were just never told what it was for”	—
16	0:23–0:25	The technique	Say the 4-step pipeline out loud once	—
17	0:25–0:27	The model, tiny	The ratio (ML section << signal processing) is the point. Framing B wins because it's the only data you have	—
18–20	0:27–0:31	Why projects fail	Slide 19 is a set piece. Do the 500-asset × 1% × 3-hrs arithmetic. Watch the number get uncomfortable. Slide 20: threshold is capacity planning, not stats	—
21–22	0:31–0:33	Control + SIL	Slide 22 is the darkest slide. <i>Slow completely down.</i> “You cannot assign a SIL rating to a neural network.” Pause	—
23	0:33–0:39	Demo one · Bearing Fault Lab	Switch to bearing-lab. Healthy (~0.3) → Early defect: ask them to find it in the raw spectrum, let them fail, then show envelope (~3.1) → Advanced (~7.6) → Noisy plant (~1.6, back to Normal, false negative)	bearing-lab
24–28	0:39–0:50	Part 4 · Digital twins	Slide 24: taxonomy + “you built one this morning.” Slide 25: the vendor test — tell them to write it down. Slide 26: extrapolation trap + kicker (“your validation set is also normal, $R^2 = 0.99$, the metric that tells you it works is computed on data that can't”). Slide 27: switch to live-monitor — frame as twin from first sentence, hold to your own test. Threshold slider: 2→8 shows warning vs false-alarms, no free lunch. Slide 28: Pi rig → model normal, measure distance	live-monitor
29–31	0:50–0:54	Part 5 · What to learn	Slide 29 “photograph this” — statistics is #1. Slide 30: Safety Instrumentation Systems vs AI/ML Healthcare are both OE2, they must choose — take safety. Slide 31: CWRU dataset project	—
32	0:54–0:56	Career close	2 min. Scarce person = knows the physical process AND the software. That's them in 2 yrs and they don't know it's valuable	—

SLIDES	TIME	TITLE	WHAT YOU DO	BROWSER
33–34	0:56–1:00	Cheat sheet + Q&A	Cheat sheet is for photos, not narration. Slide 34: both sims are theirs, ~400 lines	live-monitor

FOUR MOMENTS THE WHOLE TALK HINGES ON

1 · SLIDE 13 The BPFO derivation Rushing kills the “we knew the frequency before we measured” moment, and everything after loses foundation.	2 · SLIDE 15 Envelope from scratch VR20 stripped Signals & Systems and doesn't teach Hilbert. This <i>is</i> the value-add. Don't say “you know AM.”
3 · SLIDE 23 Early defect on bearing-lab Wait. Let them fail to see the fault in the raw spectrum. Rushing here kills the pedagogy.	4 · SLIDE 25 The vendor test The sentence that survives the talk. Tell them to write it down. Slow down for four beats.

COMPRESSIBLE IF RUNNING LONG

- §2.5 — sampling / resolution detail (~90 s)
- §3.4 — “change the speed” slider (~30 s)
- §4.4 — last threshold-slider paragraph (~1 min)

Never cut: Part 2.2 derivation · Slide 22 SIL · The vendor test · The extrapolation kicker.

Q&A — 8 ANSWERS ALREADY PREPPED IN THE SCRIPT

Will AI replace instrumentation engineers? No. Someone has to decide where the sensor goes and what happens when it's wrong. Why not deep learning on raw signal? You'd need failure data nobody has, and you can't explain it to a maintenance manager. Is my curriculum outdated? Signal processing and control aren't. The software/data layer is the gap — 40 hrs of work. How much math? Linear algebra + probability, solidly. Enough calculus for optimization. No measure theory. Twin for a small plant? Start with the cheapest thing that changes a decision. Usually 3 well-placed sensors, not a twin.	You were negative about twins. <i>Expect this.</i> Negative about surrogates trained on healthy data. Not about physics-model twins — demo two is one. LLMs in industry? Useful for unstructured text (maintenance logs, procedures). Not for vibration signals. You haven't worked in a plant. <i>Expect this — take head-on.</i> Check my arithmetic. Organizational patterns transfer. Anything specific to commissioning, ask someone else. If stumped: “I don't know. That's outside what I've actually done.” Then move on.
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VERIFY TONIGHT

Open `kiranic.com/lecture/live-monitor.html` on your presentation laptop and let it run for 10 min. Confirm MTR-402 crosses into **ALERT**, then **FAULT**. If it doesn't, fix it before Monday — not at the podium.